

INTERACTION TERMS

June 2025



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INTERACTION TERMS

- Relationship between the response and predictor varies with the value of another predictor
- Hierarchy principle: main effects needed when interaction effect is present
 - Main effect: the explanatory variable on its own
 - Interaction effect: the product of two explanatory variables



CASE STUDY

Is there a relationship between economic growth and energy consumption?

- Data from Goshrey et al. (2018)
- Variables
 - **year** Calendar year
 - **GDP** US gross domestic product
 - **EC** Energy consumption in billion BTU
 - **O_P** Oil price in dollars
 - **SPE** Public spending in dollars



THE MODEL

- Ideally, control for spending and oil price
- Energy consumption and oil price may interact to influence economic growth
- Greater volatility in oil prices after 1972, may interact with energy consumption



CREATING A VARIABLE

```
1 enrg <- enrg |>
2   mutate(V = ifelse(year < 1973,0,1))
```

year	GDP	EC	O_P	SPE	V
1949	1843106	23188436	26.74	432.94	0
1950	2004258	25106134	26.46	480.39	0
1951	2159241	26819987	24.53	483.66	0
1952	2241885	26263244	23.98	625.62	0
1953	2345289	26890640	25.03	681.04	0
1954	2330315	25603986	25.59	682.63	0
1955	2498163	28148704	25.11	667.53	0
1956	2547630	28985768	24.88	675.01	0
1957	2598811	28829838	25.67	707.75	0
1958	2575495	28328432	23.89	743.33	0
1959	2760158	29491536	23.65	794.77	0
1960	2828487	30283316	22.64	813.60	0

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year	GDP	EC	O_P	SPE	V
1961	2894416	30518655	21.91	876.43	0
1962	3069798	31724745	21.65	889.13	0
1963	3203983	32936752	21.85	920.55	0
1964	3389372	34400240	22.23	966.70	0
1965	3607118	35741681	21.94	973.36	0
1966	3842025	37556447	21.95	1055.54	0
1967	3939076	38514158	21.50	1174.94	0
1968	4129831	40616981	20.98	1259.31	0
1969	4258104	42240340	20.81	1281.86	0
1970	4266275	43320117	20.08	1323.55	0
1971	4409552	43771010	20.43	1389.60	0
1972	4643752	45636567	19.79	1457.70	0
1973	4912813	47727882	24.39	1464.24	1
1974	4885800	46086014	43.51	1477.99	1
1975	4875413	43690503	52.11	1640.30	1
1976	5136921	46103412	52.92	1747.84	1

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year	GDP	EC	O_P	SPE	V
1977	5373131	47195081	54.58	1769.93	1
1978	5672714	48011002	52.70	1817.95	1
1979	5850103	48932066	78.73	1849.23	1
1980	5834034	46393941	104.49	1969.10	1
1981	5982087	44648906	90.49	2014.03	1
1982	5865944	41990653	75.86	2128.46	1
1983	6130934	41537931	67.12	2228.35	1
1984	6571535	43794477	63.62	2265.17	1
1985	6843432	43229965	57.51	2430.01	1
1986	7080483	43527362	30.26	2530.62	1
1987	7307046	45137388	35.90	2566.27	1
1988	7607347	47082738	28.94	2644.24	1
1989	7879250	47310564	33.97	2754.87	1
1990	8027078	47437317	40.67	2893.13	1
1991	8008251	46824896	34.09	2983.27	1
1992	8279989	48075049	31.53	3069.66	1



year	GDP	EC	O_P	SPE	V
1993	8516222	48413807	26.66	3094.71	1
1994	8863097	49661516	24.27	3138.65	1
1995	9085919	50549628	25.26	3231.35	1
1996	9425866	52043051	29.96	3272.77	1
1997	9845869	52630334	26.69	3327.26	1
1998	10274695	52315561	16.80	3418.49	1
1999	10770692	52837121	22.79	3518.91	1
2000	11216350	53525625	36.54	3654.85	1
2001	11337493	52061440	29.86	3782.19	1
2002	11543143	52680407	29.12	4010.46	1
2003	11836418	52655543	34.60	4179.68	1
2004	12246868	54308755	45.78	4268.95	1
2005	12622959	53685737	58.83	4397.10	1
2006	12958525	53957409	66.45	4550.80	1
2007	13206318	54332015	71.03	4634.67	1
2008	13161982	52432125	97.33	4888.30	1



year	GDP	EC	O_P	SPE	V
2009	12757974	49802370	57.18	5415.80	1
2010	13062916	51590122	75.05	5333.77	1



REGRESSION MODEL: CATEGORICAL INTERACTION TERM

$$GDP_i = \beta_0 + \beta_1 EC_i + \beta_2 V_i + \beta_3 EC_i * V_i + \epsilon_i$$

- V_i is an indicator variable
- Before 1972 has a different slope then after 1972



WHEN $V_i = 1$

$$GDP_i = \beta_0 + \beta_1 EC_i + \beta_2 V_i + \beta_3 EC_i * 1 + \epsilon_i$$

$$GDP_i = \beta_0 + (\beta_1 EC_i + \beta_3 EC_i) + \beta_2 V_i + \epsilon_i$$

$$GDP_i = \beta_0 + (\beta_1 + \beta_3) EC_i + \beta_2 V_i + \epsilon_i$$



WHEN $V_i = 0$

$$GDP_i = \beta_0 + \beta_1 EC_i + \beta_2 V_i + \beta_3 EC_i * 0 + \epsilon_i$$

$$GDP_i = \beta_0 + \beta_1 EC_i + \beta_2 V_i + \epsilon_i$$



FITTING THE MODEL IN R

```
1 enrg.int <- lm(GDP~EC*V,data=enrg)
2 summary(enrg.int)
```

Call:

```
lm(formula = GDP ~ EC * V, data = enrg)
```

Residuals:

Min	1Q	Median	3Q	Max
-3128341	-191826	30559	351937	3423147

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.019e+06	1.270e+06	-0.802	0.42594
EC	1.259e-01	3.809e-02	3.306	0.00163 **
V	-2.070e+07	2.886e+06	-7.173	1.50e-09 ***
EC:V	4.977e-01	6.524e-02	7.629	2.57e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

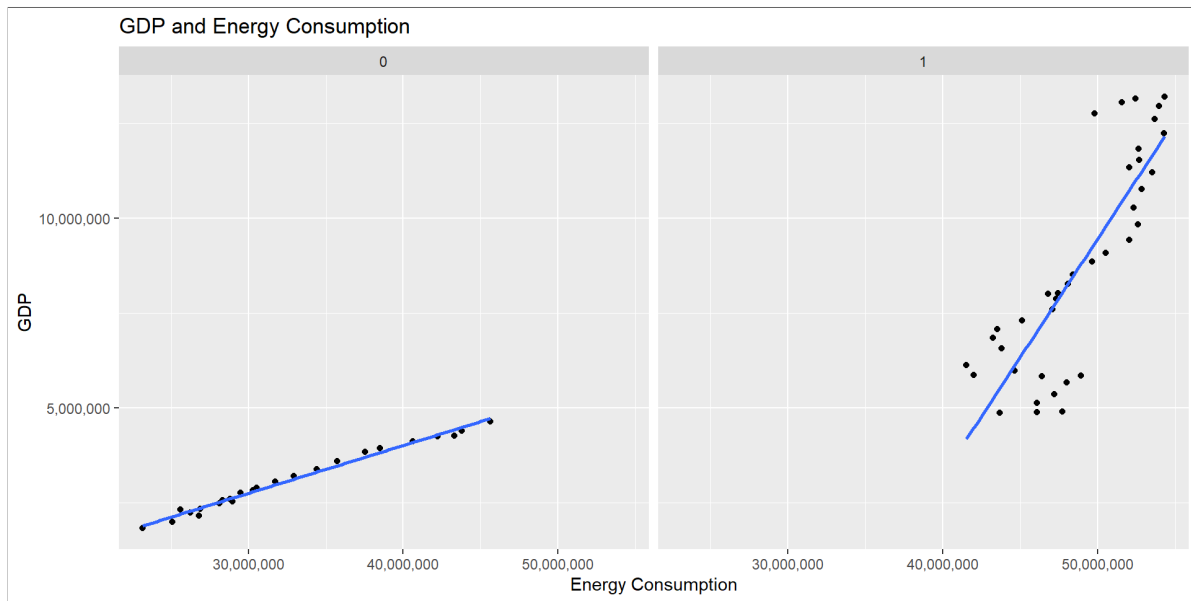


VISUALIZING THE INTERACTION: CODE

```
1 ggplot(enrg,aes(EC,GDP))+
2   geom_point() +
3   facet_wrap(vars(V)) +
4   geom_smooth(method="lm",se=FALSE) +
5   labs(title="GDP and Energy Consumption",
6        x = "Energy Consumption",
7        y = "GDP") +
8   scale_x_continuous(labels = label_comma()) +
9   scale_y_continuous(labels = label_comma())
```



VISUALIZING THE CODE: PLOTS



CODING TIME!

Using the data from the most recent practical, fit a model with a categorical interaction term.

If you are feeling ambitious, plot it!



REGRESSION MODEL: QUANTITATIVE INTERACTION TERM

$$GDP_i = \beta_0 + \beta_1 EC_i + \beta_2 OP_i + \beta_3 EC_i * OP_i + \epsilon_i$$

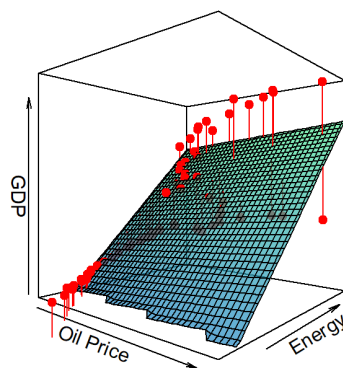
- No simplification
- How do we interpret?



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VISUALIZE: NO INTERACTION



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FITTING THE MODEL IN R

```
1 enrg.int.cont <- lm(GDP~EC*O_P,data=enrg)
2 summary(enrg.int.cont)
```

Call:

```
lm(formula = GDP ~ EC * O_P, data = enrg)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3592325	-827933	345182	878108	3341580

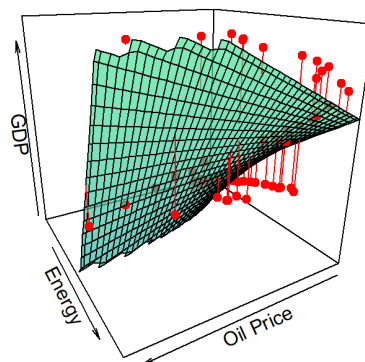
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.041e+06	2.617e+06	1.162	0.250
EC	9.227e-02	5.791e-02	1.593	0.117
O_P	-3.793e+05	9.022e+04	-4.204	9.17e-05 ***
EC:O_P	8.242e-03	1.899e-03	4.340	5.78e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



VISUALIZE: INTERACTION



EXCLUDING MAIN EFFECTS

```
1 mod.only <- lm(GDP~EC:O_P,data=enrg)
2 summary(mod.only)
```

Call:

```
lm(formula = GDP ~ EC:O_P, data = enrg)
```

Residuals:

Min	1Q	Median	3Q	Max
-6313653	-2226235	-934631	2720795	5331016

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.464e+06	6.693e+05	5.176	2.77e-06	***
EC:O_P	1.791e-03	3.249e-04	5.514	7.83e-07	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2921000 on 60 degrees of freedom



CODING TIME

Using the data from the most recent practical, fit a model with a quantitative interaction term.

Using the data from the most recent practical, fit a model with only an interaction term



SIDE NOTE

Bad residual vs fitted plot

